Day 3

Grounded Generation & Citation

AI Clinical Decision Support Lite Hackathon

Generate from evidence → **Cite every claim** → **Refuse when insufficient**

What students will learn and execute during Day 3

Today's learning objectives

Students should leave with a safe answer generator that uses only retrieved evidence.

Conceptual goal

Understand that the LLM is not the source of truth. The retrieved guideline evidence is the source of truth.

Technical goal

Build a strict grounded prompt, structured output, citation formatting, and refusal behavior when evidence is missing.

Execution goal

Connect retrieval results to generation and verify that every recommendation is traceable to document, section, and page.

Day 2 handoff check

Generation should not start until retrieval is transparent and measurable.

Required inputs

- Working Top-K retrieval.
- Retrieved chunk text.
- Document name.
- Page number.
- Section title.
- Retrieval score.

Quality baseline

The team should already have:

- 15–20 test questions
- Precision@K values
- examples of strong and weak retrieval
- selected Top-K setting

If not ready

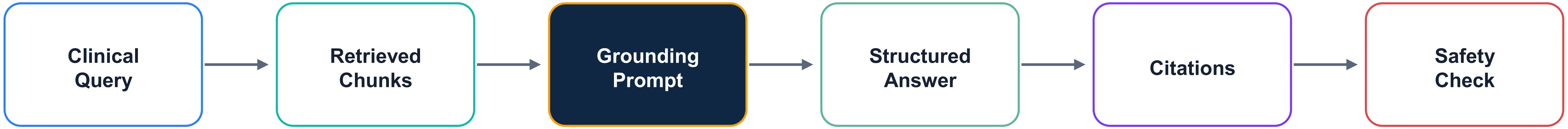
Do not rely on a strong prompt to fix bad evidence.

Fix retrieval first.

Grounded generation only works when the context is trustworthy.

Generation pipeline for today

Turn retrieved evidence into a structured answer without adding unsupported claims.



Prompt boundary

The prompt must explicitly forbid using external knowledge beyond retrieved context.

Citation binding

Each major claim should point to a chunk with document, section, and page.

Refusal path

When context is weak, the answer should say evidence is insufficient.

Grounding principle

The answer must be derived from retrieved guideline text, not from the model's memory.

INSTANT

Allowed

- Summarize retrieved chunks.
- Compare evidence across chunks.
- Quote short excerpts.
- Cite document + section/page.

Not allowed

- Add new clinical advice.
- Infer patient-specific treatment.
- Invent missing thresholds.
- Cite chunks that do not support the claim.

When unsure

State uncertainty clearly and ask for clinician review or more evidence.

In clinical RAG, a fluent unsupported answer is worse than a careful refusal.

Step 1 — Design the system prompt

The system prompt is the safety contract between retrieval and generation.

System prompt must say:

You are an evidence-grounded clinical decision support assistant.

Use only the retrieved guideline context.

If the context does not support the answer, state that evidence is insufficient.

Do not provide patient-specific diagnosis or treatment.

Every recommendation must include citations.

Keep it strict

Avoid vague instructions like “be accurate.” Use explicit rules and refusal conditions.

Keep it testable

The output should make it easy to check whether every claim has evidence.

Keep it clinical-safe

The assistant supports clinicians; it does not replace medical judgment.

Step 2 — Structure the answer

A predictable answer format makes grounding easier to judge.

Recommendation

Short, direct, and only based on retrieved chunks. No patient-specific treatment.

Supporting Evidence

Bullet points mapped to the exact evidence used. Include short excerpts when useful.

Citations

Document name + section + page. Citation should point to the supporting chunk.

Confidence & Safety

High / Medium / Low / Insufficient Evidence plus clinical disclaimer.

Step 3 — Citation formatting

Citations are not decoration. They prove traceability to the guideline.

Minimum citation

Document name
Section title
Page number

Example:
NICE Hypertension Guideline, “Starting treatment”, p.18

Better citation

Also include:
- chunk ID
- source URL if available
- retrieval score
- exact excerpt used

Bad citation

A citation is bad if it points to a page that does not actually support the claim.

Judges may check this manually.

Step 4 — Connect claims to evidence

Every clinical statement should be traceable to one or more retrieved chunks.

Generated claim

“Treatment should be considered when blood pressure remains above the guideline threshold after assessment.”

→

Supporting evidence

Chunk text excerpt
+ document name
+ section title
+ page number
+ retrieval score

If you cannot draw this link, the claim should be removed or softened.

Step 5 — Refusal behavior

A safe system must know when not to answer.

Refuse when

- No relevant chunks are retrieved.
- Retrieval scores are below threshold.
- Evidence is not specific enough.
- User asks for diagnosis or dosage for a real patient.

Good refusal

“The retrieved guidelines do not provide sufficient evidence to answer this question reliably. Please consult the relevant clinical guideline or a qualified clinician.”

Still be helpful

Mention what evidence was found, what is missing, and what type of guideline would be needed.

Step 6 — Add a confidence level

Confidence should reflect retrieval and grounding quality, not model self-belief.

Retrieval score → Evidence match → Citation coverage → Safety checks

Suggested labels

High • Medium • Low • Insufficient Evidence

Avoid exact percentages unless you have a clear calculation behind them.

Hands-on lab — Build the answer layer

Practical checklist for teams after the teaching block.

- Write the strict system prompt
- Define the structured answer schema
- Generate answers from retrieved chunks only
- Attach citations to every recommendation
- Add insufficient-evidence refusal path
- Add confidence labels
- Test direct, ambiguous, and out-of-scope questions
- Manually verify that citations support claims

Stop point: Do not ship an answer that cannot be traced back to retrieved text.

Common generation failure modes

Instructors should help students detect these issues before the final demo.

Unsupported claim

The answer includes a medical statement that is not present in the retrieved chunks.

Fix: remove, cite, or refuse.

Citation mismatch

The citation exists, but the cited page does not support the sentence.

Fix: map claims to exact chunks.

Overconfident tone

The system answers as if it is giving direct clinical advice.

Fix: use support language and disclaimer.

End-of-day review questions

Use these questions to decide whether the team is ready for Day 4: Safety & Evaluation.

Grounding quality

- Does the answer use only retrieved context?
- Are unsupported claims removed?
- Does it refuse weak evidence?

Citation quality

- Does each recommendation have a citation?
- Are document, section, and page shown?
- Do citations actually support the claims?

Safety quality

- Is there a disclaimer?
- Is confidence shown carefully?
- Are patient-specific requests handled safely?

Ready for Day 4 when the team can prove answer → citation → exact retrieved evidence.